

Project: LogiTrack

Supply Chain & Logistics Management System

Domain: Manufacturing & Logistics

1. Introduction

LogiTrack is a web-based supply chain and logistics management system designed for manufacturers, 3PL providers, and distribution enterprises to manage shipments, freight bookings, warehouse operations, supplier relationships, and delivery performance analytics. The system enables logistics coordinators to plan and track shipments, warehouse staff to manage inventory and picking operations, compliance teams to manage documentation and customs, and administrators to configure routes, carriers, and operational parameters.

It uses a REST API-based backend architecture with Angular/React frontend and is compatible with Java (Spring Boot) and .NET (ASP.NET Core) platforms.

Actors / Users

- **Shipper / Client:** Places freight orders and tracks shipment status
- **Logistics Coordinator:** Plans routes, assigns carriers, and manages dispatch
- **Warehouse Operator:** Manages inbound/outbound operations, picking, and stock
- **Driver / Field Agent:** Updates delivery status and proof of delivery
- **Compliance Officer:** Manages customs documentation and regulatory filings
- **Operations Analyst:** Monitors delivery SLAs, exceptions, and performance KPIs
- **Logistics Admin:** Configures carriers, routes, rate cards, and system parameters

2. Module Overview

- **2.1 Identity & Access Management:** Authentication, RBAC, and audit trails.
- **2.2 Shipment & Freight Management:** Manages freight orders, bookings, and shipment lifecycle.
- **2.3 Route & Carrier Management:** Configures lanes, assigns carriers, and optimizes routes.

- **2.4 Warehouse & Inventory Operations:** Manages receiving, storage, picking, and dispatch.
- **2.5 Supplier & Purchase Order Management:** Handles supplier profiles, POs, and inbound goods tracking.
- **2.6 Compliance & Documentation:** Manages customs filings, BOLs, and regulatory documents.
- **2.7 Logistics Analytics & Reporting:** Provides dashboards for delivery performance, costs, and SLAs.
- **2.8 Notifications & Alerts:** Shipment status updates, delay alerts, and compliance notifications.

3. Architecture Overview

- **Frontend:** Angular or React
- **Backend:** REST API-based architecture
- **Database:** Relational Database (MySQL / PostgreSQL / SQL Server)

4. Module-Wise Design

4.1 Identity & Access Management Module

Features

- User registration, login, and role-based access control
- Session management and audit logging

Entities

- **User**
 - UserID
 - Name
 - Role (Shipper/Coordinator/WarehouseOps/Driver/Compliance/Analyst/Admin)
 - Email
 - Phone
 - HubID
 - Status (Active/Inactive)

- **AuditLog**
 - AuditID
 - UserID
 - Action
 - EntityType
 - Timestamp

4.2 Shipment & Freight Management Module

Features

- Create freight orders and manage end-to-end shipment lifecycle
- Track shipment stages from pickup to delivery

Entities

- **FreightOrder**
 - FreightOrderID
 - ShipperID
 - OriginLocationID
 - DestinationLocationID
 - CargoDescription
 - Weight
 - Volume
 - RequiredDeliveryDate
 - Status (Draft/Booked/InTransit/Delivered/Cancelled)
- **Shipment**
 - ShipmentID
 - FreightOrderID
 - CarrierID
 - VehicleID
 - DriverID

- DispatchDate
- EstimatedArrival
- ActualArrival
- Status (Dispatched/InTransit/Delayed/Delivered/Exception)
- **DeliveryEvent**
 - EventID
 - ShipmentID
 - EventType (Pickup/InTransit/Arrived/Delivered/Exception)
 - Timestamp
 - LocationID
 - Notes

4.3 Route & Carrier Management Module

Features

- Define shipping lanes, stops, and transit times
- Assign carriers and manage rate cards

Entities

- **Route**
 - RouteID
 - OriginHubID
 - DestinationHubID
 - TransitDays
 - Mode (Road/Rail/Air/Sea)
 - Status (Active/Inactive)
- **Carrier**
 - CarrierID
 - Name
 - Mode

- ServiceLevel (Standard/Express/Overnight)
- ContactDetails
- Status (Active/Suspended)
- **RateCard**
 - RateCardID
 - CarrierID
 - RouteID
 - BaseRate
 - WeightSlab
 - EffectiveDate
 - ExpiryDate
 - Status

4.4 Warehouse & Inventory Operations Module

Features

- Manage inbound receipts, put-away, picking, and outbound dispatch
- Track bin-level inventory and movement logs

Entities

- **WarehouseInventory**
 - InventoryID
 - SKU
 - ProductName
 - WarehouseID
 - BinLocation
 - QuantityOnHand
 - QuantityReserved
 - LastUpdated
- **InboundReceipt**

- ReceiptID
- SupplierOrderID
- WarehouseID
- ReceivedDate
- ReceivedBy
- Status (Pending/Received/Discrepancy)
- **PickList**
 - PickListID
 - FreightOrderID
 - WarehouseID
 - AssignedTo
 - Status (Open/InProgress/Completed)
 - CreatedDate

4.5 Supplier & Purchase Order Management Module

Features

- Manage supplier profiles and purchase orders
- Track inbound goods and receipt confirmations

Entities

- **Supplier**
 - SupplierID
 - Name
 - Category
 - ContactDetails
 - LeadTimeDays
 - Status (Active/Inactive)
- **PurchaseOrder**
 - POID

- SupplierID
- WarehouseID
- LineItems
- TotalValue
- OrderDate
- ExpectedDelivery
- Status (Draft/Placed/PartiallyReceived/Received/Cancelled)

4.6 Compliance & Documentation Module

Features

- Generate and manage Bill of Lading, customs declarations, and regulatory filings
- Track document submission status and compliance flags

Entities

- **ShipmentDocument**
 - DocumentID
 - ShipmentID
 - DocumentType (BOL/CommercialInvoice/PackingList/CustomsDeclaration)
 - FilePath
 - SubmittedDate
 - Status (Pending/Submitted/Approved/Rejected)
- **ComplianceFlag**
 - FlagID
 - ShipmentID
 - FlagType
 - Severity
 - RaisedDate
 - Status (Open/Resolved)

4.7 Logistics Analytics & Reporting Module

Features

- Delivery performance dashboards with on-time vs. late breakdowns
- Freight cost analysis, carrier scorecards, and SLA reports

Entities

- **LogisticsReport**
 - ReportID
 - Scope (Carrier/Route/Period/Hub)
 - Metrics (ShipmentCount, OnTimeRate, AvgTransitDays, FreightCost, ExceptionRate)
 - GeneratedDate

4.8 Notifications & Alerts Module

Features

- Shipment status notifications for shippers and coordinators
- Delay alerts, exception flags, and compliance deadline reminders

Entities

- **Notification**
 - NotificationID
 - UserID
 - Message
 - Category (Shipment/Warehouse/Compliance/Carrier)
 - Status (Unread/Read/Dismissed)
 - CreatedDate

5. Deployment Strategy

- Local: Angular/React frontend, Spring Boot/.NET Core backend, local DB
- Production: Cloud or on-premises with secure APIs and stateless services

6. Database Design

Tables:

- **User**(UserID, Name, Role, Email, Phone, HubID, Status)
- **AuditLog**(AuditID, UserID, Action, EntityType, Timestamp)
- **FreightOrder**(FreightOrderID, ShipperID, OriginLocationID, DestinationLocationID, CargoDescription, Weight, Volume, RequiredDeliveryDate, Status)
- **Shipment**(ShipmentID, FreightOrderID, CarrierID, VehicleID, DriverID, DispatchDate, EstimatedArrival, ActualArrival, Status)
- **DeliveryEvent**(EventID, ShipmentID, EventType, Timestamp, LocationID, Notes)
- **Route**(RouteID, OriginHubID, DestinationHubID, TransitDays, Mode, Status)
- **Carrier**(CarrierID, Name, Mode, ServiceLevel, ContactDetails, Status)
- **RateCard**(RateCardID, CarrierID, RouteID, BaseRate, WeightSlab, EffectiveDate, ExpiryDate, Status)
- **WarehouseInventory**(InventoryID, SKU, ProductName, WarehouseID, BinLocation, QuantityOnHand, QuantityReserved, LastUpdated)
- **InboundReceipt**(ReceiptID, SupplierOrderID, WarehouseID, ReceivedDate, ReceivedBy, Status)
- **PickList**(PickListID, FreightOrderID, WarehouseID, AssignedTo, Status, CreatedDate)
- **Supplier**(SupplierID, Name, Category, ContactDetails, LeadTimeDays, Status)
- **PurchaseOrder**(POID, SupplierID, WarehouseID, LineItems, TotalValue, OrderDate, ExpectedDelivery, Status)
- **ShipmentDocument**(DocumentID, ShipmentID, DocumentType, FilePath, SubmittedDate, Status)
- **ComplianceFlag**(FlagID, ShipmentID, FlagType, Severity, RaisedDate, Status)
- **LogisticsReport**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Shipper Portal**: Freight order placement, shipment tracking, document download
- **Coordinator Dashboard**: Dispatch planning, carrier assignment, route map view

- **Warehouse Console:** Inbound receipt, bin inventory, pick list management
- **Driver/Field App:** Active delivery tasks, status updates, proof of delivery capture
- **Compliance Portal:** Document checklist, customs filing status, compliance flag tracker
- **Operations Dashboard:** SLA monitoring, delivery performance KPIs, exception queue
- **Admin Console:** Carrier management, route configuration, rate card setup

8. Non-Functional Requirements

- **Performance:** Handle 50,000+ shipment events per day with sub-second tracking updates
- **Security:** Role-based data access, encrypted document storage, and full audit trails
- **Scalability:** Supports multi-hub and cross-border logistics expansion
- **Availability:** 99.9% uptime with disaster recovery for critical dispatch operations
- **Maintainability:** Modular services, API versioning, and configurable SLA rule engine
- **Observability:** Real-time shipment dashboards, carrier scorecards, and exception alerts

9. Assumptions & Constraints

- Phase 1 excludes integration with external GPS telematics and ERP systems
- Route optimization is manual; algorithmic optimization is deferred to Phase 2
- All notifications are in-app only; SMS/email integrations are out of Phase 1 scope
- Customs clearance workflow is tracked internally; integration with customs portals is deferred
- Fully implementable using standard Java Spring Boot / ASP.NET Core, Angular/React, and RDBMS tooling